

|   |   |   |   |
|---|---|---|---|
| 0 | 3 | . | 1 |
|---|---|---|---|

Describe **and** explain the relationship between surface area to volume ratio of the human body and metabolic rate.

**[3 marks]**

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**Question 3 continues on the next page**

**Table 2** shows the height and mass of two adults.

**Table 2**

| Person | Height / cm | Mass / kg |
|--------|-------------|-----------|
| A      | 181         | 90.90     |
| B      | 149         | 62.62     |

The surface area of a person is estimated using the following formula:

$$\text{Surface area in m}^2 = \sqrt{\frac{\text{height in cm} \times \text{mass in kg}}{3600}}$$

The volume of a person is estimated using the following formula:

$$\text{Volume in m}^3 = \frac{\text{mass in kg}}{1010}$$

0 3 . 2

Using suitable calculations, deduce which person has the smaller surface area to volume ratio.

Show your working **and** complete the sentence below.

Give your answer to 3 significant figures.

**[3 marks]**

Person \_\_\_\_\_ has the smaller surface area to volume ratio which = \_\_\_\_\_

0 3 . 3

Which is **not** a possible explanation for the difference in surface area to volume ratio between person **A** and person **B**?

Tick (✓) **one** box.

[1 mark]

Person **A** and person **B** have different body shapes.

☐

Person **A** has a higher mass than person **B**.

☐

Person **A** is taller than person **B**.

☐

0 3 . 4

Name **two** structural features of the gills of a fish that increase their surface area.

[1 mark]

1 \_\_\_\_\_

\_\_\_\_\_

2 \_\_\_\_\_

\_\_\_\_\_

Turn over for the next question

Turn over ►

